(Please write your Enrollment Number) Enrollment No.

MID-TERM EXAMINATION

(Course Name : B. Tech, CSE/IT/CSE-AI/AI&ML/Reappear) (Semester : 4th sem)

(March, 2024) OFF LINE mode

Subject Code: BIT 202	Subject: Operating Systems
Time :1 ½Hours	Maximum Marks : 30
Note:Q. 1 is compulsory.	

Q1	All parts are compulsory	(2.5*4)	
	(a) Explain Process & Process states in detail.	UNIT-1	
	(b) Define the Terms : 1) long term Scheduler 2) Short term Scheduler	UNIT-1	
	(c) Define Segmentation with Paging	UNIT-2	
	(d) Define Deadlock and its 4 conditions	UNIT-2	

Q2 (Attempt any Two Parts) UNIT-1 **(5,5)**

(a) Consider the set of 6 processes whose Arrival time and Burst time are given below :

Process ID	Arrival Time	Burst Time
P_1	5	5
P_2	4	6
P_3	3	7
P_4	1	9
P_5	2	2
P_6	6	3

If the CPU Scheduling policy is Round Robin with time quantum = 3 , Calculate the average Waiting Time and Average Turn Around Time.

(b) Consider the set of 4 processes whose Arrival time and Burst time are given below and mode is SJF with pre–emption :

Calculate the Average waiting time and Average turn–around time.

Process No.	Arrival Time	Busrt Time
P_1	0	5
P_2	1	3
P_3	2	4
P_4	4	1

(C) Consider the set of 5 processes whose Arrival time and Burst time are given below.

Find the Average waiting time and Average Turn–Around time using Pre – emptive priority Scheduling.

Consider Higher number represents higher priority.

Process ID.	Priority	Arrival Time	Burst Time
P_1	2	0	4
P_2	3	1	3
P_3	4	2	1
P_4	5	3	5
P_5	5	4	2

Q3 (Attempt any Two Parts) UNIT-2 **(5,5)**

(a) Consider the Page Referenées 7,0,1,2,0,3,0,4,2,3,0,3,2,3 with 4 page frames. Find the number of Page faults using:

i) Optimal Page Replacement

ii) Least Recently Used

(b) Explain the concept of Logical Address Space and Physical Address Space.

(C) Explain Contiguous Memory Allocation